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Jharkhand University of Technology, Ranchi

Diploma 1st Semester Examination, 2025 (NEP)

Subject : Engineering Mathematics | **Subject Code : BSC 101**

Time Allowed : 3 Hours | **Full Marks : 70**

Pass Marks : 21

Answer in your own words.

Answer any five *questions in which* Question No. 1 *is compulsory.*

The figures in the margin indicate full marks.

Marks are given in the right margin.

1. Choose the correct answer in each of the following: 2×7=14

(i) If $2\begin{pmatrix}3 & 4\\ 5 & x\end{pmatrix}+\begin{pmatrix}1 & 2y\\ 0 & 1\end{pmatrix}=\begin{pmatrix}7 & 0\\ 10 & 5\end{pmatrix}$, then values of x and y are

(a) $x=8, y=0$ (b) $x=2, y=-4$

(c) $x=6, y=2$ (d) $x=5, y=3$

(ii) If $\begin{pmatrix}2x+y\\ x-y\end{pmatrix}=\begin{pmatrix}8\\ 4\end{pmatrix}$, then values of x and y are

(a) $x=4, y=0$ (b) $x=8, y=0$

(c) $x=6, y=2$ (d) $x=5, y=3$

(iii) The distance between the points $A(2,-7)$ and $B(6,3)$ is

(a) 4 units (b) 6 units

(c) 8 units (d) 10 units

(iv) The distance between the points $A(2,-3)$ and $B(-6,3)$ is

(a) $\sqrt{29}$ units (b) $-\sqrt{29}$ units

(c) $4\sqrt{29}$ units (d) $2\sqrt{29}$ units

(v) If the slope of the line joining the points $A(x,2)$ and $B(6,-8)$ is $\frac{-3}{4}$, then x =?

(a) $\frac{-22}{3}$ (b) $\frac{22}{3}$

(c) $\frac{-2}{3}$ (d) 3

(vi) $\sin\left(\frac{19\pi}{3}\right)$ =?

(a) $\frac{\sqrt{3}}{2}$ (b) $-\frac{\sqrt{3}}{2}$

(c) $\frac{1}{2}$ (d) $-\frac{1}{2}$

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(vii) $\operatorname{cosec} 840° = ?$

(a) $\frac{2}{\sqrt{3}}$ (b) $\frac{\sqrt{3}}{2}$

(c) $\frac{1}{2}$ (d) $-\frac{1}{2}$

2. (a) Solve the following system of linear equations (use any method):

$$3x + 4y + 2z = 8$$
$$x - 2y - 3z = 3$$
$$3x - 2y + 6z = -2$$

(b) Find the eigenvalues and eigenvectors of the matrix $A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$. 7+7

3. (a) Find the equation of the line whose portion intercepted between the axes is bisected at the point $(3, -2)$.

(b) Find the equation of the line whose y-intercept is -3 and which is perpendicular to the line joining the points $(-2, 3)$ and $(4, -5)$. 7+7

4. (a) If $\sin\theta = \frac{\sqrt{3}}{2}$ and θ lies in quadrant II, find the values of all the other five trigonometric functions.

(b) Calculate the values of: (i) $\sin 15°$, (ii) $\cos 15°$, (iii) $\tan 15°$. 7+7

5. (a) Differentiate y $w.r.t.x$:

$$y = \log\left(\operatorname{cosec}\frac{x}{2} + \cot\frac{x}{2}\right)$$

(b) Find the point of local maxima or minima and corresponding local maximum and minimum value for the following function: 7+7

$$f(x) = -x^3 + 12x^2 - 5$$

6. (a) Evaluate the integral: $I = \int_0^{\frac{\pi}{2}} \cos^4 x dx$.

(b) Evaluate the integral: $I = \int \left(\frac{x^3+4x^2-3x-2}{x+2}\right) dx$ 7+7

7. Answer *any four* of the following: 3.5×4=14

(a) If $2\begin{pmatrix} x & 5 \\ 7 & y-3 \end{pmatrix} + \begin{pmatrix} 3 & -4 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 7 & 6 \\ 15 & 14 \end{pmatrix}$, then find the value of x and y.

(b) Find the equation of the line perpendicular to the line $x - 7y + 5 = 0$ and having x-intercept 3.

(c) Prove that $\left(\frac{\cos x+\sin x}{\cos x-\sin x} - \frac{\cos x-\sin x}{\cos x+\sin x}\right) = 2\tan 2x$.

(d) A spherical soap bubble is expanding so that its radius is increasing at the rate of 0.02 cm/sec. At what rate is the surface area increasing when its radius is 5 cm? (Take $\pi = 3.14$)

(e) Evaluate the integral: $I = \int \left(\frac{2}{1+x^2} - \frac{2}{\sqrt[3]{x^4}} + 3x^{-1} - \frac{3}{x\sqrt{x^2-1}} + 4e^{\frac{x}{4}} + 5^x\right) dx$

Jharkhand University of Technology, Ranchi

Diploma 1st Semester Examination, 2025 (NEP)

Subject : Mining Geology-I

Time Allowed : 3 Hours

Subject Code : MIN-101

Full Marks : 70

Pass Marks : 21

Answer in your own words.
Answer any five questions in which Question No. 1 is compulsory.
The figures in the margin indicate full marks.
All questions carry equal marks.

1. Choose the correct option: 2×7

(i) The weathering caused by the temperature changes is known as
(a) Chemical weathering
(b) Physical weathering
(c) Biological weathering
(d) None of these

(ii) The process of breaking rocks by plant roots is
(a) Oxidation
(b) Leaching
(c) Frostation
(d) Biological weathering

(iii) The removal and transportation of weathered material is called
(a) Weathering
(b) Deposition
(c) Erosion
(d) Sedimentation

(iv) Limestone is an example of
(a) Igneous Rock
(b) Sedimentary Rock
(c) Metamorphic Rock
(d) Magmatic Rock

(v) Rocks formed under heat and pressure are known as
(a) Sedimentary
(b) Igneous
(c) Metamorphic
(d) Volcanic

(vi) The process of formation of igneous rocks is called
(a) Crystallization
(b) Diagenesis
(c) Lithification
(d) None of these

(vii) The upper most solid part of the earth where mining activities occur is
(a) Inner core
(b) Outer core
(c) Lithosphere
(d) Mental

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2. (a) Explain the scope and importance of geology in mining engineering. 7+7

   (b) Describe any four branches of geology.

3. (a) Describe the various internal constitution of the earth. 7+7

   (b) Explain the process of weathering and its type.

4. (a) Discuss in detail, the classification of rocks. 7+7

   (b) Explain the physical properties of Calcite, Gypsum, Talc and Magnesite.

5. (a) Write the various characteristics of igneous rocks. 7+7

   (b) Explain the different types of metamorphism.

6. (a) Differentiate between weathering and erosion. 7+7

   (b) Describe the origin and occurrence of mineral.

7. Write short notes on *any four*: 3.5×4

   (a) Hydrogeology

   (b) Ultra basic

   (c) Dolerite

   (d) Sandstone

   (e) Quartzite

Jharkhand University of Technology, Ranchi

Diploma 1st Semester Examination, 2025 (NEP)

Subject : Computer Aided Engineering Drawing

Subject Code : CSE-103

Time Allowed : 3 Hours

Full Marks : 70

Pass Marks : 21

Answer in your own words.

Answer any five *questions in which* Question No. 1 *is compulsory.*

The figures in the margin indicate full marks.

All questions carry equal marks.

1. Answer the following multiple-choice questions: 2×7=14

(i) Which instrument is used for measuring angles?

(a) Compass (b) Divider

(c) Protractor (d) Scale

(ii) Which type of line is used for dimension lines?

(a) Thick continuous (b) Thin continuous

(c) Chain thin (d) Zigzag

(iii) Which scale enlarges dimensions?

(a) Full scale (b) Reduced scale

(c) Enlarged scale (d) Natural scale

(iv) Which dimensioning system places dimensions in one direction only?

(a) Aligned (b) Unidirectional

(c) Chain (d) Parallel

(v) Which symbol represents Third Angle projection?

(a) Cone symbol (b) Truncated cone symbol

(c) Circle symbol (d) Square symbol

(vi) Which projection gives a pictorial view?

(a) Orthographic (b) Isometric

(c) Perspective (d) Oblique

(vii) Which CAD command is used to set drawing limits?

(a) LIMITS (b) GRID

(c) SNAP (d) UNITS

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2. (a) Explain conventions of lines and their applications. 7+
   (b) Draw a polygon using representative fraction and scale.

3. (a) Explain principle planes of projection with neat sketch. 7+
   (b) Draw projection of a line perpendicular to HP and parallel to VP.

4. (a) Draw projection of a square plane inclined to HP and perpendicular to VP. 7+7
   (b) Draw projection of a circle resting on edge and inclined to both planes.

5. (a) Differentiate pictorial view between orthographic view with suitable example. 7+7
   (b) Explain difference between perspective and isometric projection.

6. (a) List hardware requirements for CAD and explain their importance. 7+7
   (b) Draw a polygon and ellipse using CAD commands and apply constraints.

7. Write short notes on *any four* of the following: 3.5×4=14
   (a) Representative fraction and its use
   (b) Types of scales in engineering drawing
   (c) Symbols of First and Third Angle projection
   (d) Basic CAD plotting parameters
   (e) Difference between 2D and 3D CAD modeling

Jharkhand University of Technology, Ranchi

Diploma 1st Semester Examination, 2025 (NEP)

Subject : Communication Skill

Time Allowed : 3 Hours

Subject Code : BSC 104

Full Marks : 70

Pass Marks : 21

Answer in your own words.
Answer any five questions in which Question No. 1 is compulsory.
The figures in the margin indicate full marks.
All questions carry equal marks.

1. Choose the correct answers: 2×7

(i) The word 'lip' is a CVC word because
(a) it has a blend of two consonants
(b) it has three consonants
(c) it follows the consonant–vowel–consonant pattern.
(d) It has two vowels

(ii) The feminine gender of 'lion' is
(a) She-lion
(b) Tigress
(c) Leopardess
(d) Lioness

(iii) Choose the correct plural of 'Mouse'.
(a) Mices
(b) Mous
(c) Mice
(d) Mouses

(iv) 'The gardener waters the plants'. (Change into Passive form)
(a) The plants watered by the gardener.
(b) The plants are watered by the gardener.
(c) The plants were watered by the gardener.
(d) The plants have been watered by the gardener.

(v) The body language is a part of
(a) non-verbal communication.
(b) verbal communication.
(c) written communication.
(d) visual communication.

(vi) Which of the following is required to sign in to Gmail?

(a) Mobile number only (b) Username and password

(c) Only password (d) None of these

(vii) Which skill is MOST important in telephonic communication?

(a) Reading (b) Drawing

(c) Listening (d) Writing

2. (a) Explain in detail, the eight parts of speech with suitable examples.

(b) Explain various barriers to effective communication. 7+7

3. (a) Describe the different types of verbal communication.

(b) Discuss the significance of body language in communication. 7+7

4. (a) Define the art of professional writing. Discuss its importance in academic and technical fields.

(b) Define context. Explain the importance of context. 7+7

5. (a) Discuss the importance of communication tools.

(b) Write an email to your friend Suman Kumar/Anjali Kumari telling him/her about your recent trip to Netarhat, Jharkhand. 7+7

6. (a) Discuss in detail the different types of sentences.

(b) Write a short paragraph on 'The first day in college' (100 words). 7+7

7. Write short notes on the following: 3·5×4

(a) Webinar

(b) Curriculum Vitae

(c) PowerPoint presentation

(d) Gesture
